The thermomechanical response: what moves the wall, and how long it takes

A lag-and-gain screen of every candidate driver, against the on-structure sensors, ERA5 and the Gubbio station

Study report · `studies/03_thermomechanical_response/`

Abstract

Study 01 established what the monitoring record contains and Study 02 characterised the external sources that describe the weather at the site. Neither measured how the wall itself responds. This report screens every candidate driver the project has against one response—the compensated, cleaned inclination—in three rounds, one for each source family: the sensors on the structure, the ERA5 reanalysis, and the meteorological station in Gubbio town. Each pair is scanned over the operators its physics admits—a transport delay and a thermal time constant, jointly, because a filter and a shift can produce the same peak correlation while meaning different things—and fitted at the operator the scan selects, on the levels and on the diurnal band separately, with Newey–West standard errors and a negative control carried through every table. Air temperature is the strongest driver in every round and its association is negative, as the site geometry predicts; the response follows the on-structure air temperature with no resolvable delay at hourly resolution, follows solar radiation by one hour, and *leads* the wall-temperature probe by two. The screen is reported with its own noise floor, which turns out not to be silent, and with the time constants it could not identify, which turn out to be most of them.

Contents

1 Introduction

A grey-box model of this structure cannot be specified until three things are known about each candidate driver: whether the wall responds to it at all, after how long, and in which direction. This report measures those three quantities.

The question is not new to the project, but the answers on record are not usable. Section 7.5 of `docs/raw-data-format.md` states the expectation that the site geometry imposes—the valley face of the embankment is the more exposed, so daytime heating expands it and tips the wall towards the mountain, which is a *negative* change by the instrument's convention—and records against it a measured positive slope on the raw channel, on two different units of hardware. Both statements come from studies now retired to `studies/obsolete/`, whose conclusions this project does not build upon.

This study therefore re-measures the coupling on trusted ground: the response and the on-structure drivers come from Study 01's exported archive, the external drivers are read through the same loaders and channel maps Study 02 uses, and every function that computes anything lives in `studies/shmlib/`.

What this report does not do. It does not re-open the raw-versus-compensated contradiction of Section 7.5. The response screened here is `inc_comp_cleaned`, Study 01's analysis column, and nothing else. The contradiction remains recorded where it is recorded; the numbers below are measured on the compensated series and describe that series.

2 Method

2.1 The response

`inc_comp_cleaned` on the hourly UTC grid. Three of Study 01's decisions are consumed and none is revisited: the compensation, the join across the February 2025 instrument change, and the spike verdict, which marks the samples Study 01 replaced by interpolation. Those samples are dropped before any hour is averaged. An interpolated value is the interpolator's output rather than the instrument's, and a coupling measured against one would be partly a coupling to Study 01's cleaning.

The absolute level of this column means nothing—it is fixed by how the instrument sat in its mount and by the anchors Study 01 applied—so only its changes carry structural information.

2.2 The two bands

Every series is screened twice: as its level, and as its diurnal band, defined as the series minus its 24-hour centred rolling mean. The two answer different questions. A correlation on the levels is dominated by whatever slow variation the two series share; a correlation on the band is a statement about the daily forcing cycle alone. The filter is one line of arithmetic rather than a designed band-pass, so that its definition is reproducible and no unstated phase response sits between the data and the lags measured afterwards. Figure 1 shows what the separation does to one week of the response.

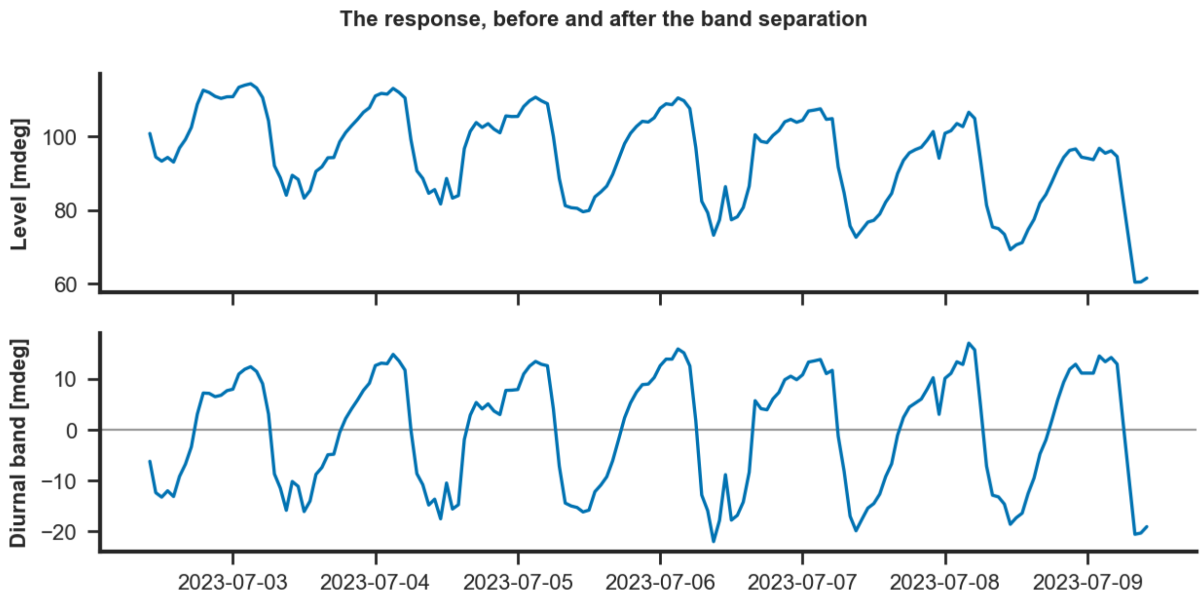

Figure 1: The response over one week, as recorded and after the band separation. The upper panel carries the level, whose position on the axis is set by an arbitrary anchor; the lower panel carries the diurnal band, in which only the daily excursion survives.

2.3 Two things can delay a response, and they are not the same thing

A *transport delay* shifts a driver in time without changing its shape: it is the time a thermal front takes to travel from the exposed face to the depth that governs the movement. A *thermal inertia* low-passes it: masonry does not follow the air around it, it integrates it, and the discrete form of that lumped-capacitance response is a one-pole filter with a time constant τ.

The two are not interchangeable, and a scan over delay alone cannot tell them apart: a filter with the right time constant and no delay, and a delay with no filter, produce very similar peak correlations while implying entirely different physics. Every pair in this report is therefore scanned over both, and the whole grid is kept, because the maximum of such a grid is usually not a peak but a ridge—delay and time constant substitute for one another over a wide range, and a single winning cell quoted alone claims a resolution the record does not have.

Two restrictions on that grid are load-bearing.

The diurnal band is scanned over delay alone. On a band holding essentially one period the two parameters are not separately identifiable at all. A one-pole filter applied to a signal of one frequency is exactly an amplitude scaling and a phase shift, and a delay is also a phase shift, so every combination on a line of constant total phase fits equally well. Scanned jointly there, the search rotates the phase past a half cycle and reports the flipped sign as its optimum: the two external air temperatures, which stand at $r = -0.81$ and -0.79 against the response, came back as $+0.82$ at a delay of five hours and τ of eighteen. That is an artefact of fitting an unidentifiable parameter, not a finding, and pinning the band's grid to $\tau = 0$ removes it.

On the levels the grid stops at three days, by a stated prior. The mass whose temperature governs the inclination is a wall face and its effective time constant is hours to a couple of days. Left unbounded the scan does not settle on anything of that order: run out to 2160 h, the optimum for the external drivers slides to between 720 and 1440 h—a month to two months—at which point the filtered driver is no longer weather but season, and what it correlates with is the response's own annual drift. The negative control slides the same way. Section 7 reports which time constants survived that bound and which did not.

2.4 Lags, and the bound on them

Each pair is scanned over integer-hour delays, the driver shifted forward so that a positive lag means the response follows. The scan maximises $|r|$ rather than r: the expected association here is negative, and a search maximising the signed correlation would walk away from the optimum it was sent to find.

External forcings are scanned over 0 to 12 hours. Both bounds are physical. A forcing must precede the response it causes, so a negative optimum against one is not a measurement of anything. Twelve hours is where a delay stops being distinguishable from a lead: on a near-periodic daily signal a delay of d hours and a lead of $24 - d$ produce the same alignment, and a lead cannot be represented by a causal filter or used by a forecast.

Wall temperature is exempt and is scanned over ± 24 hours. It is not a forcing but an internal state variable at an unknown depth, and nothing requires it to precede the deformation: a shallow probe would be led by the wall's motion and a deep one would follow it.

2.5 Gains, and their uncertainty

At the selected operator—both the delay and the time constant—the response is regressed on the driver by ordinary least squares, and the slope is reported in millidegrees per unit of the driver. Fitting against the operator that won, rather than against the driver as recorded, is not a detail: on a synthetic pair built with a known gain of -2.5 and a six-hour time constant, regressing on the unfiltered driver returns -0.78. Its standard error is Newey–West with a one-day truncation. This is not a refinement: the residuals of an hourly geophysical regression are strongly autocorrelated, and the ordinary standard error of such a fit understates the uncertainty by enough to make a coincidence look significant.

2.6 The strata, and the negative control

Every scan is run over the whole record, over each instrument era, and over each season, on the season definitions Studies 01 and 02 use. Supply voltage is carried through every round as a negative control. A bounded search over a near-periodic pair returns an optimum for every candidate, so what separates a coupling from that background is not the size of $|r|$ alone but its size relative to a channel that has no mechanism by which it could move a wall. Section 10 reports what the control actually did, which is not what was expected of it.

3 What the record supports

The three source families do not cover the same record. The on-structure air temperature, humidity and supply voltage run the length of the post-outage archive; solar radiation and wall temperature exist only in the current instrument era and are further reduced by Study 01's verdicts, leaving roughly five thousand and roughly forty-eight hundred paired hours respectively against some twenty-one thousand for air temperature. Every result below is reported with the paired hours it rests on, and the two current-era channels are read in that light.

Driver	Hours	Paired	First	Last
`batt_str`	21,776	21,161	2023-06-21	2026-08-19
`rh_str`	21,776	21,161	2023-06-21	2026-08-19
`sr_str`	5,256	5,102	2025-02-20	2026-08-19
`tair_str`	21,776	21,161	2023-06-21	2026-08-19
`twall_str`	4,903	4,754	2025-02-20	2026-08-19
`tair_gs`	27,095	20,563	2023-06-21	2026-08-12
`rh_gs`	27,052	20,534	2023-06-21	2026-08-12
`tdew_gs`	27,096	20,567	2023-06-21	2026-08-12
`wspd_gs`	27,036	20,510	2023-06-21	2026-08-12
`wdir_gs`	27,119	20,586	2023-06-21	2026-08-12
`pres_gs`	26,788	20,343	2023-06-21	2026-08-12
`rain_gs`	27,119	20,586	2023-06-21	2026-08-12
`rain_rate_gs`	27,118	20,586	2023-06-21	2026-08-12
`sr_gs`	27,118	20,585	2023-06-21	2026-08-12
`tair_era5`	27,394	20,855	2023-06-21	2026-08-12
`rh_era5`	27,394	20,855	2023-06-21	2026-08-12
`tdew_era5`	27,394	20,855	2023-06-21	2026-08-12
`sr_era5`	27,394	20,855	2023-06-21	2026-08-12
`rain_era5`	27,394	20,855	2023-06-21	2026-08-12
`wspd_era5`	27,394	20,855	2023-06-21	2026-08-12
`wdir_era5`	27,394	20,855	2023-06-21	2026-08-12
`wdir_sin_gs`	27,119	20,586	2023-06-21	2026-08-12
`wdir_cos_gs`	27,119	20,586	2023-06-21	2026-08-12
`wdir_sin_era5`	27,394	20,855	2023-06-21	2026-08-12
`wdir_cos_era5`	27,394	20,855	2023-06-21	2026-08-12

Table 1: What each driver has to work with: the hours it carries, the hours it shares with the response, and the extent of that overlap.

4 Round 1 · the on-structure drivers

The channels measured at the wall are the closest thing to a local forcing the project has, and each carries a caveat Study 01 established: the air temperature is recorded inside a sun-exposed housing rather than a meteorological screen, the radiation exists only on the current-era days Study 01 did not condemn, and the wall-temperature probe is absent on more than half the days of that era.

On the diurnal band the housing air temperature is the strongest association in the whole study, $r = -0.957$ at zero lag, with a gain of -2.79 mdeg/°C. Solar radiation follows at $r = -0.867$ with a delay of one hour and a gain of -0.026 mdeg per watt per square metre. Both signs are the sign the site's geometry predicts, and neither needs an operator: their time constants are zero and the instantaneous fit is what the winning cell reproduces.

On the levels the on-structure radiation behaves differently. Its instantaneous fit explains almost nothing, $R^2 = 0.17$, and a 48-hour time constant lifts that to 0.29 — $\Delta R^2 = 0.122$, the largest operator gain of any driver whose optimum is not against a bound. Read with Section 7, that is a statement about the slow band being dominated by variation neither the wall nor the radiation resolves, rather than a measurement of the wall's thermal mass.

The wall temperature is the interesting case. On the diurnal band its optimum sits at -2 hours and on the levels at -4 hours with a time constant of four, and both say the same thing: the inclination *leads* the probe. Under the reading of Section 7.5 that is the signature of a probe sitting deep enough that the layers actually driving the movement—those nearest the sun-exposed valley face—reach their extremes before it does. It is a measurement, not an error.

It is also the one driver whose operator is worth what it costs. On the band the delay buys $\Delta R^2 = 0.214$ over the instantaneous fit; on the levels the delay and the time constant together buy 0.071, and both parameters land inside their grids rather than against a bound, so this is the only external driver in the study whose thermal time constant the record actually identifies.

What the clamp costs can now be stated rather than asserted. Restricting the search to non-negative delays—the restriction any forecast is under, since a lead would need the probe's future values at the moment of issue—gives up $\Delta R^2 = 0.068$ on the band and 0.024 on the levels. That is the price of using this channel predictively, and it is small enough to pay.

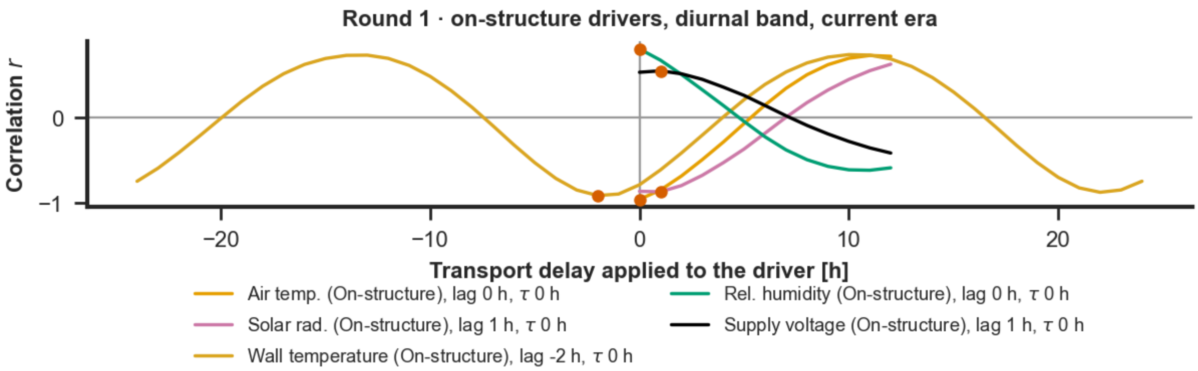

Figure 2: Round 1. Correlation against lag for each on-structure driver, diurnal band, current era. The marked point on each curve is the optimum the scan selected. The supply voltage is drawn with the rest: it is the negative control, and its curve is the floor the others are read against.

5 Round 2 · ERA5

ERA5 is a nine-kilometre grid average: its air temperature is not the air temperature at the wall and its radiation is not the radiation the wall receives. What it offers instead is a record with no gaps across the whole archive, which is precisely what the on-structure channels lack.

On the diurnal band its air temperature reaches $r = -0.787$ at zero lag with a gain of -2.04 mdeg/°C, and its radiation $r = -0.873$, also at zero lag. Both are weaker than their on-structure counterparts and both keep the expected sign. The attenuation is what a spatial average should do to a local signal, and it is consistent with Study 02's finding that the three sources are not replicates of one quantity.

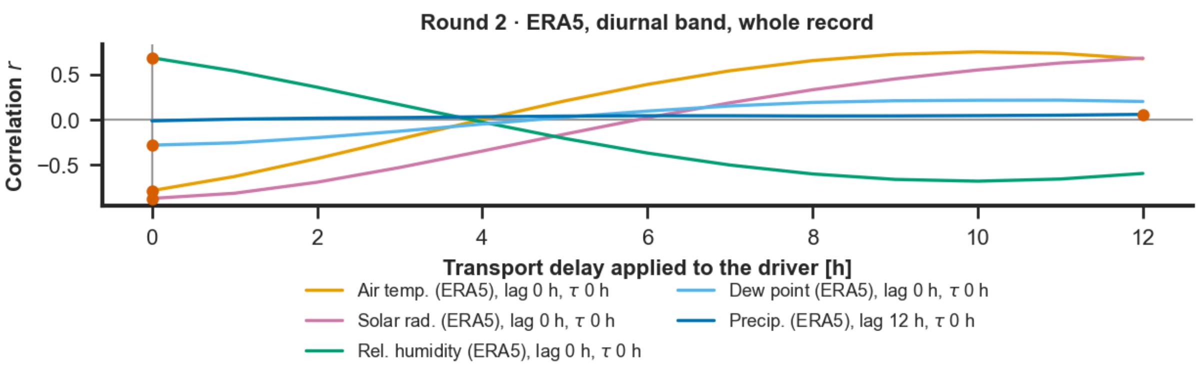

Figure 3: Round 2. Correlation against lag for the ERA5 channels, diurnal band, whole record.

6 Round 3 · the town station

The Gubbio station is a point observation from a standard screen, neither a grid average nor a box bolted to the wall. On the diurnal band its radiation gives the strongest external association of the study, $r = -0.879$ at zero lag with a gain of -0.035 mdeg per watt per square metre, and its air temperature $r = -0.807$ with a gain of -2.23 mdeg/°C. Both sit between the on-structure channels and ERA5, which is where a town-centre screen ought to sit.

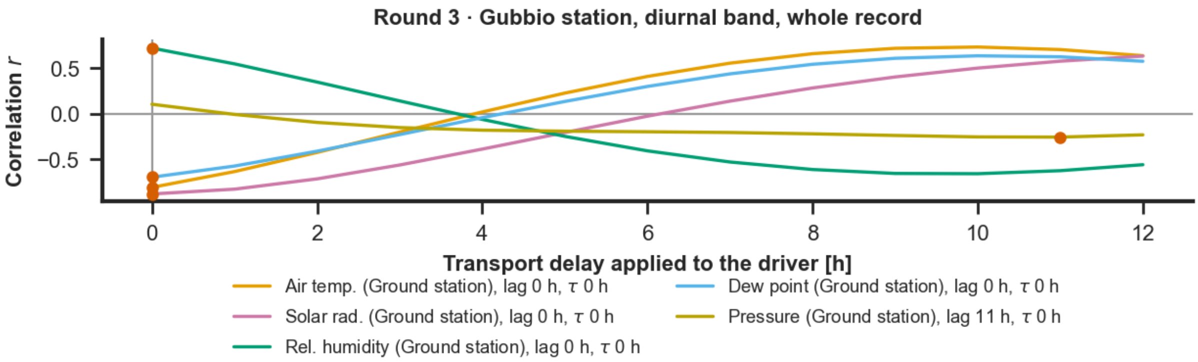

Figure 4: Round 3. Correlation against lag for the Gubbio station channels, diurnal band, whole record.

7 What the operator scan settled, and what it did not

Scanning delay and time constant jointly was meant to answer whether the wall's response is delayed, damped, or neither. It answered that for two drivers and refused to answer it for the rest, and the refusal is the more useful half.

Two time constants are identified. The wall temperature settles at $\tau = 4$ h on the levels and the on-structure radiation at $\tau = 48$ h, both strictly inside the grid, both buying real explanatory power over the instantaneous fit ($\Delta R^2 = 0.071$ and 0.122). The radiation's placement is the weaker of the two: its surface is nearly flat from 36 to 72 h ($R^2 = 0.285, 0.288, 0.287$), so what that panel establishes is that the coupling is inertial rather than instantaneous, not that the time constant is two days. On the diurnal band every driver's time constant is zero by construction, and the delays are zero or one hour, so the daily coupling is instantaneous at hourly resolution.

Every other time constant on the levels is unidentified. Fifteen of the twenty-two drivers pin against the three-day bound rather than settling inside it, which means the scan stopped where the grid did and not where the physics did. Figures 6 and 7 show what that looks like: ten panels, ten surfaces climbing monotonically to the top row, not one of them turning over inside the grid. Extending the grid does not fix this: run to 2160 h the optima slide to 720–1440 h, where a "filtered driver" is an annual cycle and its correlation with the response is a statement about season. The negative control does exactly the same, $r = -0.47$ at the bound, which is the tell—a channel with no mechanism cannot acquire one by being smoothed, so what the smoothing found is variation the whole site shares.

The consequence for the report is a restriction on what the levels are allowed to say. On that band the study reports the instantaneous association and the two identified operators, and treats a bound-pinned time constant as missing rather than as measured. The diurnal band carries the coupling result.

This also corrects the source of the idea. The retired study in `studies/obsolete/inclination_predicti` whose figures prompted this scan, capped its own grid at 168 h and reported radiation gaining

$\Delta R^2 = 0.43$ from a time constant sitting exactly on that cap. That is the same slide, unremarked. The method it demonstrates is sound and this report adopts it; the number it produced is an artefact of where its grid ended.

7.1 How to read one of these grids

The three figures that follow are the study's only two-dimensional ones and they carry most of what the tables cannot. This subsection says how to read them.

What is drawn. Each cell is one operator applied to the driver before the driver is regressed on the response, and the two axes are the operator's two parameters. They stand for the two distinct things a wall can do to a forcing before the inclinometer sees it: answer late, or answer slowly.

The horizontal axis is the *transport delay*, in hours. It shifts the driver in time without changing its shape: at a delay of d hours the response now is paired with the driver as it was d hours earlier. Physically it is travel time. Heat entering the exposed face needs time to reach the depth whose expansion governs the inclination, and the thermal front arrives there with its shape intact but late. A delay is what a forcing looks like after crossing a distance.

The vertical axis is the *thermal time constant* τ, in hours. It does not shift the driver; it smooths and damps it. The driver is passed through a first-order low-pass filter, the discrete form of a lumped thermal mass: each hour the filtered value moves towards the current driver by the fraction $\Delta t/(\tau + \Delta t)$ of the distance still between them, so that a step in the driver is answered by an exponential approach that has covered about 63 % of its way after τ hours and 95 % after 3τ. A body of time constant τ does not follow the forcing around it; it integrates it, with a memory of about τ hours, and its temperature is a rounded copy of the forcing whose fast swings are damped and whose peak arrives later the larger τ is. A time constant of zero returns the driver untouched. The filter is applied first and the delay second, following the physics: the mass integrates the forcing, and the resulting thermal state then takes time to reach the depth that moves the instrument.

The two are easy to confuse on a single sinusoid and separable on a real record, which is why the grid scans them jointly rather than one at a time. On a pure 24-hour cycle a delay of d hours and a time constant whose phase lag is d hours are the same phase shift, and no correlation can tell them apart. What separates them is that a delay moves every frequency by the same time, whereas the filter lags and damps the fast components more than the slow ones: on the diurnal cycle with its harmonics, the multi-day weather swings and the sharp events such as a cold front or a clear day after cloud, a delayed driver stays sharp where a filtered one is blurred, and the regression can tell which shape the response follows. A scan over delay alone reads the two as one number; the grid reads them as a pair.

The colour of a cell is the R^2 of the driver, put through that delay and that time constant, regressed on the response over the hours the two series share at that operator. Dark is a high R^2, following the project's rule that the meaningful end of a scalar map is the dark one. The dot marks the winning cell and the panel title repeats its coordinates and its R^2. The bottom-left cell, $(0, 0)$, is the instantaneous case: no delay, no inertia, the driver exactly as recorded. Moving right along the bottom row asks whether the wall answers late but sharply; moving up the left column asks whether it answers on time but blurred; the interior asks both at once.

Two things the panels do not say, and one they do. The vertical axis is a list of chosen time constants drawn at equal spacing, not a linear scale, so a step from 48 to 72 h occupies the same height as one from 0 to 1 h and vertical distance is not proportional to hours. And R^2 discards the sign: a panel cannot distinguish a driver that heats the wall from one that cools it, and the sign must be read from the r column of Table 2 or 3.

What the panels do say, and this is deliberate, is how the drivers compare. All three figures are drawn on one colour scale, running from 0 to the largest R^2 any of their panels reaches, rounded up—0.90, set by the on-structure air temperature. A shade therefore means the same number in every panel of every one of the three figures, and nothing is clipped. The price is that a weak panel occupies only the bottom of the range and shows little of its own structure: the station's pressure is a uniform pale field because it explains between 0.6 and 1.0 per cent of the variance everywhere on its grid, and that flatness is the finding rather than a defect of the drawing.

The shapes, and what each one means. A grid is read by its shape before its maximum, because the shape says whether the maximum means anything.

- **A bright corner at $(0, 0)$, fading in both directions.** The coupling is instantaneous: neither a delay nor an inertia improves it. The on-structure air temperature is this, and it is the cleanest result in the study.
- **A bright band along the bottom row, away from zero.** A pure transport delay: the response reproduces the driver's shape, later. Nothing in this study is cleanly of this kind, which is itself informative.
- **A bright band up the left column, at zero delay.** A pure inertia: the response follows a smoothed driver rather than a shifted one. The on-structure radiation on the levels is this, rising from $R^2 = 0.166$ at $\tau = 0$ to 0.288 at $\tau = 48$ h. Note how weakly the maximum is placed even so—0.285, 0.288 and 0.287 at 36, 48 and 72 h—so the shape is established and the exact time constant is not.
- **A diagonal ridge.** The two parameters are trading off: a longer delay with a shorter time constant fits about as well as the reverse, because both mostly move the driver's phase. Only the combination is resolved, and the winning pair is one point on a ridge rather than a measurement of either parameter. Wall temperature shows this around its optimum, where the four best cells—$(-4, 4)$, $(-4, 3)$, $(-5, 4)$ and $(-5, 6)$—lie within 0.002 of each other in R^2.
- **A surface that climbs to the top row and stops there.** The optimum is the bound, not a turning point: the scan was stopped by the grid rather than by the physics, and the time constant is unidentified. All ten external panels of Figures 6 and 7 are this shape. Such a cell must not be quoted as a measured time constant, and the tables flag it.
- **Two bright regions about twenty-four hours apart.** Aliasing. On a near-periodic signal a delay of d and a delay of $d \pm 24$ produce almost the same alignment, so a scan wide enough to contain both will report whichever happens to score higher. Wall temperature, scanned over signed delays, peaks at -4 h with $R^2 = 0.802$ and again at $+19$ h with 0.777: the second is the first wearing a different sign of delay, and an unbounded scan that returned it would describe the wall as following its own probe by most of a day.
- **A pale, featureless panel.** No coupling at any operator, and on the shared scale that pallor is directly comparable with every other panel rather than relative to the panel's own contents. Station pressure is this, and so, more surprisingly, is the on-structure radiation on the levels: its interior maximum is real but it sits at $R^2 = 0.29$, against 0.89 for the air temperature beside it.

Reading a panel with the tables. The figure locates the optimum; two columns of the tables say whether it is worth anything. ΔR^2 is the winning cell's R^2 minus the instantaneous cell's, so a small value means the operator is decoration however confident the panel title looks—the on-structure air temperature's is exactly zero. For a driver scanned over signed delays, R^2 lost to causality is what disappears when the search is restricted to the non-negative half of the panel, which is the half a forecast is allowed to use.

One panel, end to end. Take the wall temperature in Figure 5. The bright region is a ridge, so the delay and the time constant are not separately sharp. Its maximum sits at a *negative* delay, so the response leads the probe, which Section 7.5 reads as a probe mounted deeper than the layers that drive the movement. There is a second bright region near +19 h, which is that maximum's daily alias and not a second mechanism. The maximum is inside the grid on both axes, so unlike every external driver this time constant is identified. Table 2 then supplies what the panel discarded: the association is negative, $r = -0.895$, the gain is -2.95 mdeg per degree, the operator is worth $\Delta R^2 = 0.071$ over the instantaneous fit, and restricting it to causal delays costs 0.024.

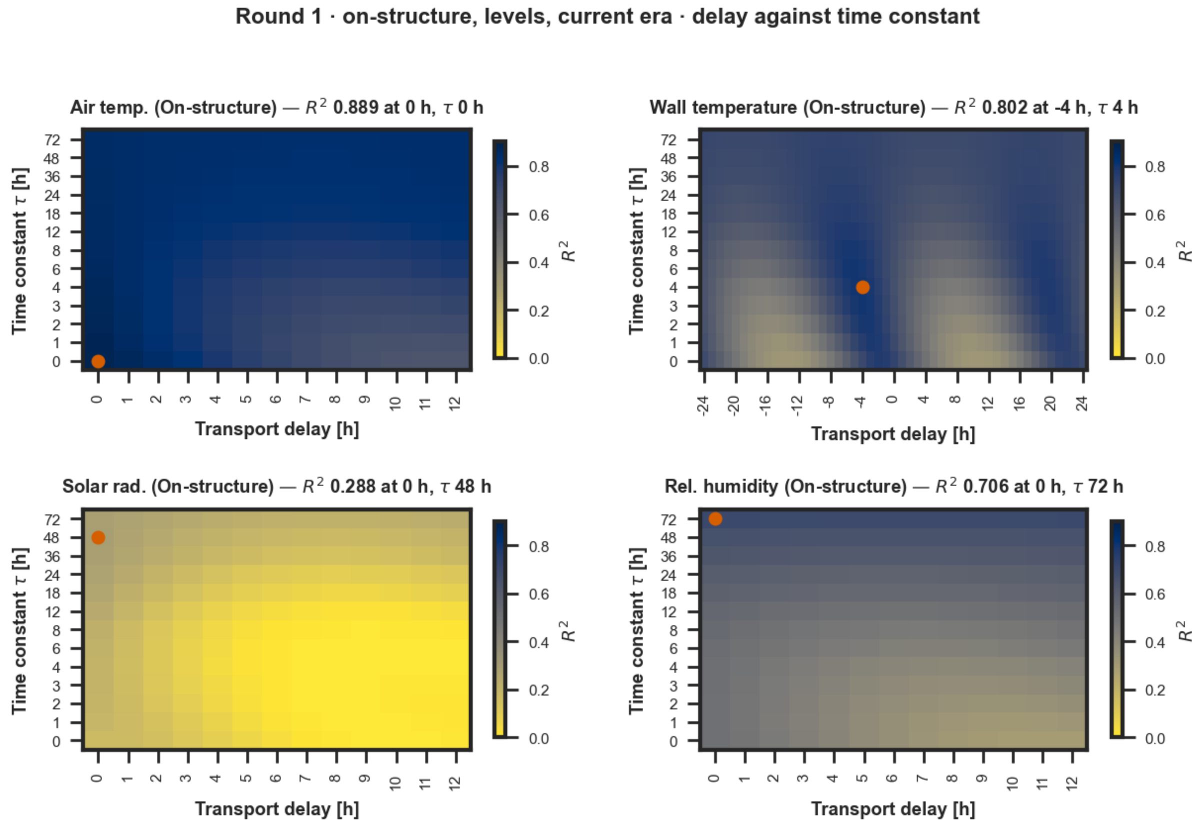

Figure 5: The delay-by-time-constant grid for the four on-structure drivers with a mechanism, on the levels, current era. Each panel is titled with its own winning cell, marked in the accent colour. All three grid figures share one colour scale, 0 to 0.90, taken from the strongest panel among them so that nothing is clipped and a shade means the same number throughout. What the table cannot show is here: where the bright region is a ridge rather than a peak, the delay and the time constant are trading off against one another and the winning pair is one point on that ridge. The wall temperature's second ridge, around a delay of twenty hours, is the 24-hour alias of its optimum at -4 and is what the signed scan's bound exists to keep out of the answer.

The same grid for the two external source families says something the on-structure one does not, and it says it at a glance.

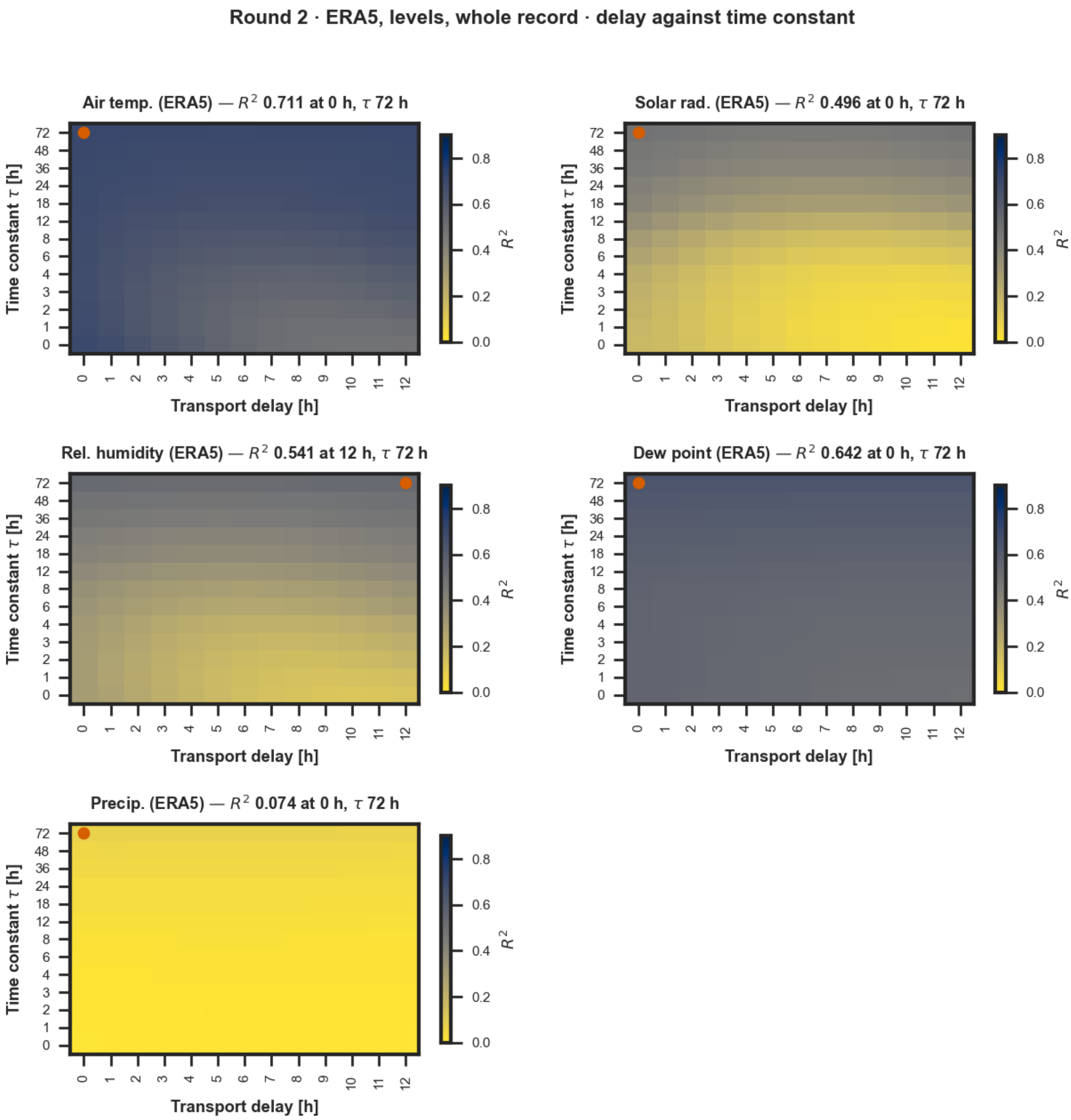

Figure 6: The grid for the ERA5 channels, on the levels, whole record. Every panel has the same shape: the surface climbs monotonically with the time constant and never turns over, so its maximum lies on the top row of the grid rather than inside it. That is the visible form of a time constant the record does not identify—the scan is not finding an optimum, it is being stopped by the bound. Compare the on-structure panels of Figure 5, where the wall temperature and the radiation have an interior maximum with structure around it.

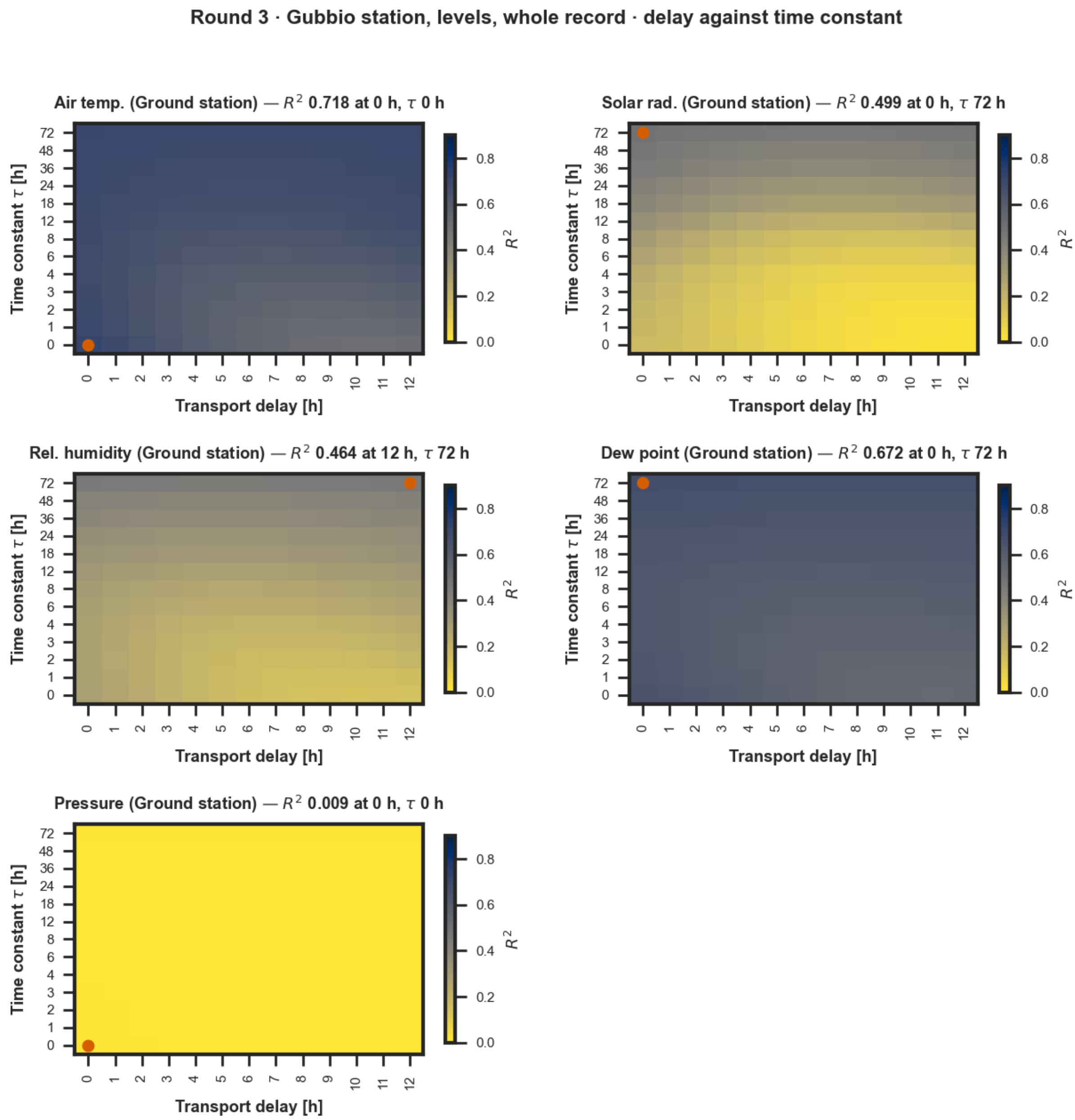

Figure 7: The grid for the Gubbio station channels, on the same terms and the same shared colour scale. The picture is the reanalysis's: five channels, five maxima on the bound. Because the scale is shared with Figure 5, the comparison between the families can be made by eye—these panels are pale where the on-structure ones are dark, and the station's pressure, coupled to nothing, is the palest field in the study.

8 Across the rounds

The point of screening three source families rather than one is that the three disagree, measurably so in Study 02, and a coupling that survives the disagreement is a statement about the wall rather than about a file.

Air temperature survives it. It is the leading or near-leading driver in every round, always at zero lag on the diurnal band, always negative, with gains of −2.79, −2.23 and −2.04 mdeg/°C for the housing, the station and ERA5 respectively. The ordering is the exposure ordering: the channel closest to the wall gives the largest gain, and the spatial average the smallest.

Solar radiation survives it too, at zero to one hour on the diurnal band, with gains between −0.026 and −0.035 mdeg per watt per square metre. On the levels all three radiation channels want a long time constant and two of the three want more than the grid allows, which Section 7 reads as a seasonal covariation rather than a thermal one.

Relative humidity appears in every round with a *positive* association of $r \approx +0.69$ to $+0.79$. This is not an independent finding. Humidity at this site is largely the inverse of the daily temperature cycle, and its correlation with the response carries the sign that inversion implies. It is reported because the screen reports what it measures, not because a moisture mechanism has been demonstrated.

The tables below give the full screen, one row per driver, pooled over the whole record.

Driver	**Lag**	τ	**r**	ΔR^2	**Slope**	**N**	**Clears**	**Sign**
`rh_str`	0	72	+0.712	+0.134	+1.83	21,161	yes	no
`sr_str`	0	48	-0.538	+0.122	-0.171	5,102	yes	yes
`tair_str`	0	0	-0.865	+0.000	-3.79	21,161	yes	yes
`twall_str`	-4	4	-0.895	+0.071	-2.95	4,727	yes	yes
`tair_era5`	0	72	-0.843	+0.008	-4.37	20,855	yes	yes
`rh_era5`	12	72	+0.735	+0.239	+2.68	20,861	yes	no
`tdew_era5`	0	72	-0.801	+0.084	-5.9	20,855	yes	yes
`sr_era5`	0	72	-0.704	+0.319	-0.286	20,855	yes	yes
`rain_era5`	0	72	+0.271	+0.067	+109	20,855	no	no
`wspd_era5`	6	72	+0.239	+0.053	+11.7	20,861	no	no
`wdir_sin_era5`	0	72	+0.076	+0.001	+9.24	20,855	no	no
`wdir_cos_era5`	12	72	-0.126	+0.013	-10.9	20,861	no	yes
`tair_gs`	0	0	-0.848	+0.000	-3.91	20,563	yes	yes
`rh_gs`	12	72	+0.681	+0.177	+2.37	20,528	yes	no
`tdew_gs`	0	72	-0.820	+0.026	-5.28	20,567	yes	yes
`wspd_gs`	8	2	+0.091	+0.008	+1.1	20,506	no	no
`pres_gs`	0	0	+0.093	+0.000	+0.473	20,343	no	no
`rain_gs`	0	72	+0.308	+0.089	+198	20,586	no	no
`rain_rate_gs`	0	36	+0.075	+0.005	+4.62	20,586	no	no
`sr_gs`	0	72	-0.706	+0.329	-0.317	20,585	yes	yes
`wdir_sin_gs`	0	72	+0.510	+0.224	+85.5	20,586	yes	no
`wdir_cos_gs`	10	72	-0.113	+0.011	-25.4	20,585	no	yes

Table 2: The screen on the levels. **Lag** and τ in hours, ΔR^2 what the operator bought over the instantaneous fit, **Slope** in millidegrees per unit of the driver, **Clears** whether the driver exceeds the negative control, **Sign** whether its slope points the way the site's geometry predicts. A τ of 72 h is the grid's bound and means the time constant was not identified; see Section 7.

Driver	Lag	τ	r	ΔR^2	Slope	N	Clears	Sign
rh_str	0	0	+0.790	+0.000	+0.686	20,204	yes	no
sr_str	1	0	-0.867	+0.010	-0.0262	5,008	yes	yes
tair_str	0	0	-0.957	+0.000	-2.79	20,204	yes	yes
twall_str	-2	0	-0.910	+0.214	-1.75	4,673	yes	yes
tair_era5	0	0	-0.787	+0.000	-2.04	19,850	yes	yes
rh_era5	0	0	+0.686	+0.000	+0.476	19,850	yes	no
tdew_era5	0	0	-0.283	+0.000	-2.17	19,850	no	yes
sr_era5	0	0	-0.873	+0.000	-0.0329	19,850	yes	yes
rain_era5	12	0	+0.059	+0.003	+1.91	19,874	no	no
wspd_era5	0	0	-0.447	+0.000	-4.87	19,850	yes	yes
wdir_sin_era5	0	0	+0.135	+0.000	+3.69	19,850	no	no
wdir_cos_era5	10	0	+0.236	+0.004	+6	19,870	no	no
tair_gs	0	0	-0.807	+0.000	-2.23	19,329	yes	yes
rh_gs	0	0	+0.722	+0.000	+0.597	19,297	yes	no
tdew_gs	0	0	-0.695	+0.000	-4.14	19,349	yes	yes
wspd_gs	10	0	+0.437	+0.061	+1.35	19,252	yes	no
pres_gs	11	0	-0.255	+0.054	-2.64	18,992	no	yes
rain_gs	0	0	+0.052	+0.000	+1.46	19,372	no	no
rain_rate_gs	0	0	+0.038	+0.000	+0.0895	19,372	no	no
sr_gs	0	0	-0.879	+0.000	-0.0353	19,371	yes	yes
wdir_sin_gs	0	0	+0.176	+0.000	+2.45	19,372	no	no
wdir_cos_gs	2	0	+0.305	+0.009	+4.45	19,365	no	no

Table 3: The same screen on the diurnal band, where the time constant is fixed at zero because it is not identifiable against a single period. The band is where the daily forcing lives, and where every association in this study is strongest.

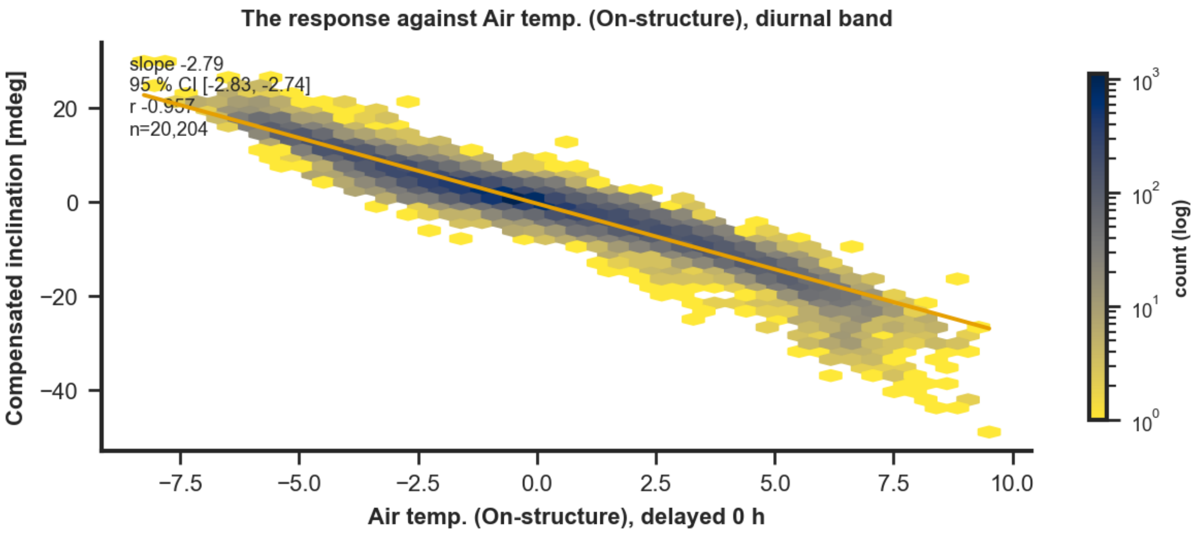

Figure 8: The response against the strongest surviving driver at the lag the scan selected, diurnal band, with the fitted gain and its interval.

9 Stability

A gain fitted over a whole record is a single number that a single season can produce on its own. Re-fitting the same lag month by month separates a coupling from a coincidence: a real one holds its slope, within its uncertainty, across months in which the driver's own range changes.

The on-structure air temperature holds a median gain of −2.48 mdeg/°C across thirty-five months, ranging between −3.63 and −1.88. The two external air temperatures behave the same way about their own medians. Solar radiation is stable to within a factor of about three across the same window. The wall-temperature and on-structure radiation gains rest on nine months only, which is the whole current era, and their stability is correspondingly weakly established.

Wind speed channels are visibly unstable: ERA5 wind speed ranges from −0.84 to −14.6 mdeg per m/s and station wind speed from 0.09 to 3.02 across months. Neither should be carried forward as a driver on the strength of a pooled fit.

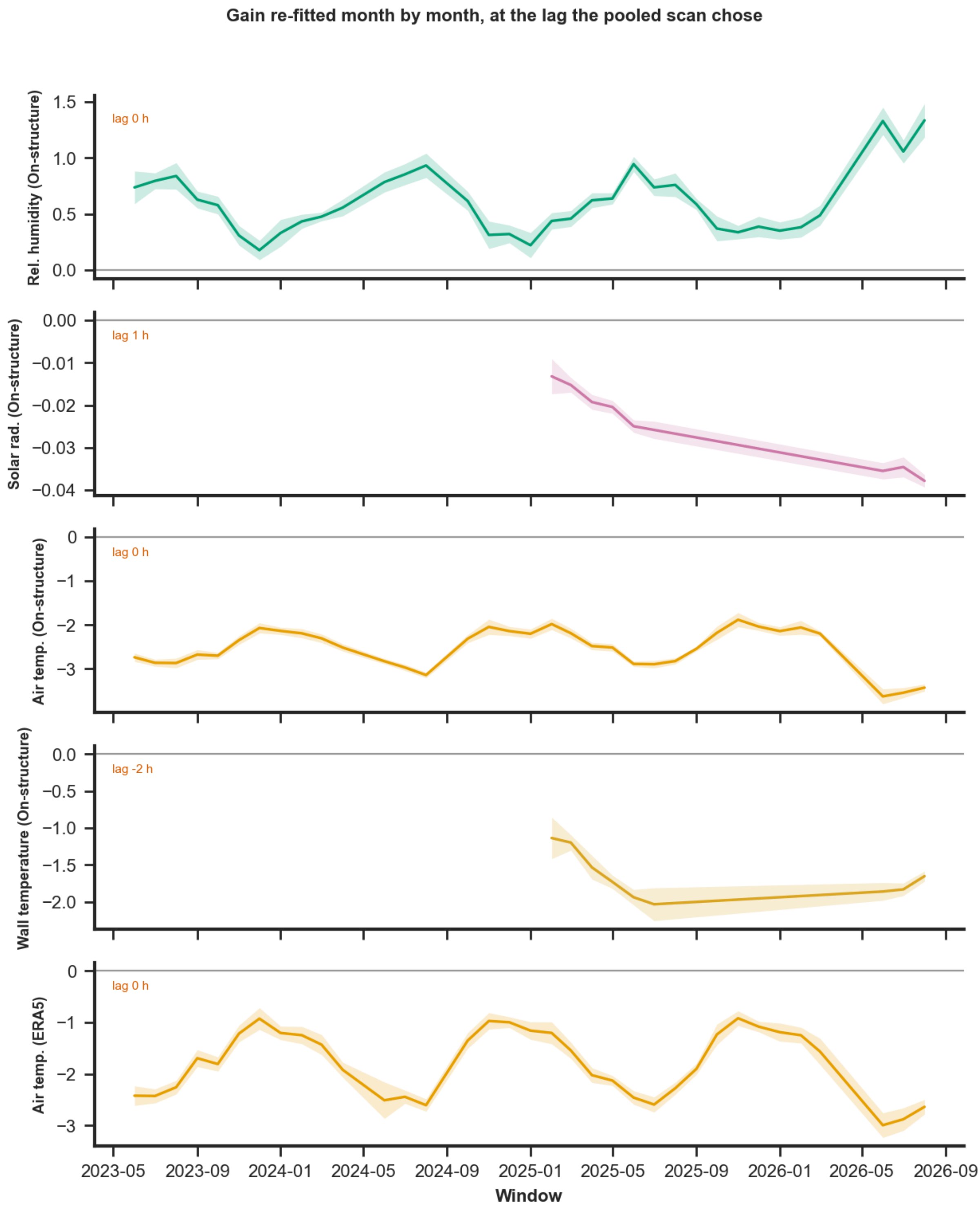

Figure 9: Gain re-fitted month by month at the lag the pooled scan selected, with its Newey–West interval. One panel per driver: the gains are in different units and an axis holding several of them would say nothing about any.

10 The control is not silent

The supply voltage was carried as a negative control on the reasoning that no mechanism moves a wall by battery voltage, so whatever it scored would be the study's noise floor. It did not behave as a null channel. On the levels over the post-outage record it reaches $r = -0.472$, and on the diurnal band it reaches -0.673 within the post-outage legacy era and $+0.549$ within the current one.

On the levels it also takes the largest time constant the grid offers, landing at $r = -0.47$ against the bound exactly as the candidate drivers do.

The explanation is not that the wall responds to the battery. It is that the battery responds to the same daily cycle as everything else—a solar-charged supply in a sun-exposed housing carries the day in its voltage—and, on the slow band, to the same annual one. The control sliding to the bound alongside the candidates is what identifies that bound as an artefact rather than a measurement, and it is the clearest thing the control did in this study. The control is therefore not mechanism-free, and the floor it sets is a *conservative* one—too high rather than too low. A driver that clears it has cleared a channel that itself carries the diurnal forcing, which makes clearing a stronger statement than it was designed to be; a driver that fails to clear it has not thereby been shown to be uncoupled.

Read that way, thirteen of the twenty-two screened drivers clear the control on the diurnal band; the channels that do not include precipitation, the wind direction components, station pressure, and ERA5 dew point.

A better control for a future study would be a channel with a comparable sampling history and no thermal exposure at all. The project does not currently record one.

11 Verdict

Band	Driver	Lag	τ	r	ΔR^2	Slope	N	Clears	Signif.	Sign
diurnal	`pres_gs`	11	0	-0.255	+0.054	-2.64	18,992	no	yes	yes
diurnal	`rain_era5`	12	0	+0.059	+0.003	+1.91	19,874	no	yes	no
diurnal	`rain_gs`	0	0	+0.052	+0.000	+1.46	19,372	no	yes	no
diurnal	`rain_rate_gs`	0	0	+0.038	+0.000	+0.0895	19,372	no	yes	no
diurnal	`rh_era5`	0	0	+0.686	+0.000	+0.476	19,850	yes	yes	no
diurnal	`rh_gs`	0	0	+0.722	+0.000	+0.597	19,297	yes	yes	no
diurnal	`rh_str`	0	0	+0.790	+0.000	+0.686	20,204	yes	yes	no
diurnal	`sr_era5`	0	0	-0.873	+0.000	-0.0329	19,850	yes	yes	yes
diurnal	`sr_gs`	0	0	-0.879	+0.000	-0.0353	19,371	yes	yes	yes
diurnal	`sr_str`	1	0	-0.867	+0.010	-0.0262	5,008	yes	yes	yes
diurnal	`tair_era5`	0	0	-0.787	+0.000	-2.04	19,850	yes	yes	yes
diurnal	`tair_gs`	0	0	-0.807	+0.000	-2.23	19,329	yes	yes	yes
diurnal	`tair_str`	0	0	-0.957	+0.000	-2.79	20,204	yes	yes	yes
diurnal	`tdew_era5`	0	0	-0.283	+0.000	-2.17	19,850	no	yes	yes
diurnal	`tdew_gs`	0	0	-0.695	+0.000	-4.14	19,349	yes	yes	yes
diurnal	`twall_str`	-2	0	-0.910	+0.214	-1.75	4,673	yes	yes	yes
diurnal	`wdir_cos_era5`	10	0	+0.236	+0.004	+6	19,870	no	yes	no
diurnal	`wdir_cos_gs`	2	0	+0.305	+0.009	+4.45	19,365	no	yes	no
diurnal	`wdir_sin_era5`	0	0	+0.135	+0.000	+3.69	19,850	no	yes	no
diurnal	`wdir_sin_gs`	0	0	+0.176	+0.000	+2.45	19,372	no	yes	no
diurnal	`wspd_era5`	0	0	-0.447	+0.000	-4.87	19,850	yes	yes	yes
diurnal	`wspd_gs`	10	0	+0.437	+0.061	+1.35	19,252	yes	yes	no
level	`pres_gs`	0	0	+0.093	+0.000	+0.473	20,343	no	yes	no
level	`rain_era5`	0	72	+0.271	+0.067	+109	20,855	no	yes	no
level	`rain_gs`	0	72	+0.308	+0.089	+198	20,586	no	yes	no
level	`rain_rate_gs`	0	36	+0.075	+0.005	+4.62	20,586	no	yes	no
level	`rh_era5`	12	72	+0.735	+0.239	+2.68	20,861	yes	yes	no
level	`rh_gs`	12	72	+0.681	+0.177	+2.37	20,528	yes	yes	no
level	`rh_str`	0	72	+0.712	+0.134	+1.83	21,161	yes	yes	no
level	`sr_era5`	0	72	-0.704	+0.319	-0.286	20,855	yes	yes	yes
level	`sr_gs`	0	72	-0.706	+0.329	-0.317	20,585	yes	yes	yes
level	`sr_str`	0	48	-0.538	+0.122	-0.171	5,102	yes	yes	yes
level	`tair_era5`	0	72	-0.843	+0.008	-4.37	20,855	yes	yes	yes
level	`tair_gs`	0	0	-0.848	+0.000	-3.91	20,563	yes	yes	yes
level	`tair_str`	0	0	-0.865	+0.000	-3.79	21,161	yes	yes	yes
level	`tdew_era5`	0	72	-0.801	+0.084	-5.9	20,855	yes	yes	yes
level	`tdew_gs`	0	72	-0.820	+0.026	-5.28	20,567	yes	yes	yes
level	`twall_str`	-4	4	-0.895	+0.071	-2.95	4,727	yes	yes	yes
level	`wdir_cos_era5`	12	72	-0.126	+0.013	-10.9	20,861	no	yes	yes
level	`wdir_cos_gs`	10	72	-0.113	+0.011	-25.4	20,585	no	yes	yes
level	`wdir_sin_era5`	0	72	+0.076	+0.001	+9.24	20,855	no	yes	no
level	`wdir_sin_gs`	0	72	+0.510	+0.224	+85.5	20,586	yes	yes	no
level	`wspd_era5`	6	72	+0.239	+0.053	+11.7	20,861	no	yes	no
level	`wspd_gs`	8	2	+0.091	+0.008	+1.1	20,506	no	yes	no

Table 4: The verdict, pooled over the whole record. The three conditions are reported separately rather than combined, because a driver can fail one and pass another and the distinction matters to what a later study may do with it.

What a later study may take from this report:

- **Air temperature is the driver.** Negative, at zero lag on the diurnal band, in all three source families, with a gain between -2.0 and -2.8 mdeg/°C that is stable over thirty-five months. Where the on-structure channel is present it is the strongest of the three; where it is absent, either external source carries the same sign and lag at a reduced gain.
- **Solar radiation couples at zero to one hour** with a gain of order -3×10^{-2} mdeg per watt per square metre, and the external radiation channels reproduce the on-structure one closely enough to stand in for it outside the certified window.
- **Wall temperature leads nothing.** The inclination leads it by two hours on the diurnal band and by four, with a four-hour time constant, on the levels. It is the only driver whose time constant this record identifies. Use the measurement as a diagnostic of probe depth; clamping the lead to zero, which any forecast requires, costs $\Delta R^2 = 0.068$ on the band and 0.024 on the levels.
- **No thermal inertia is identified for the external drivers.** Their level-band time constants pin against the grid's bound and are reported as missing. A model built on this study should carry them instantaneously, at the delays the diurnal band measured, and should not inherit a time constant from Table 2.
- **The sign agrees with the site's structural convention** for every heating driver in every round, on the compensated response. This does not resolve the contradiction of Section 7.5, which concerns the raw channel, and it must not be quoted as though it did.

12 Limitations

- **The compensated response is not an independent measurement of the structure.** Section 7.2's compensation subtracts a fixed 5 mdeg/°C from a channel whose own thermal slope is of comparable size, so part of what is measured here as a structural response is the residual of that subtraction. Screening the raw channel alongside it was deliberately left out of scope; it is the obvious next study.
- **Zero lag at hourly resolution is a bound, not a value.** A response that follows its driver by twenty minutes and one that follows it instantly are the same number on this grid. Study 01's archive is on a twenty-minute grid and could support a finer scan.
- **The levels cannot separate inertia from season.** Section 7 sets this out: fifteen of twenty-two time constants pin against the grid's bound, and the negative control pins with them. A record of three years contains only three realisations of the annual cycle, which is not enough to tell a slow thermal response from a seasonal coincidence. Nothing short of a different experiment—or a driver whose annual cycle differs from the response's—will settle it.
- **The current-era channels rest on nine months.** Wall temperature and on-structure radiation carry roughly five thousand paired hours against twenty-one thousand for air temperature, and their stability is established correspondingly weakly.
- **The negative control is thermally exposed**, as Section 10 sets out. The floor it defines is conservative and the screen inherits that.
- **Correlation between drivers is not addressed.** Air temperature, dew point, humidity and radiation share one daily cycle, and this study screens each against the response separately rather than fitting them together. Which of them a model should carry, and with what collinearity treatment, is a question for the modelling study, not for this screen.